MAT334: Complex Variables

Allison Lau

Summer 2023

Contents

1 Introduction **1**
1.1 The Field of Complex Numbers . . . 1
1.2 Properties . . . 2
1.3 Elementary Functions . . . 7

2 Basic Topology **12**
2.1 Open and closed sets . . . 12
2.2 Limits and continuity . . . 14
2.3 Connected and path-connected sets . . . 18
2.4 Compact Sets . . . 20

3 Complex Differentiation **22**
3.1 The Complex Derivative . . . 22
3.2 Conformal Maps . . . 25
3.3 Cauchy-Riemann Equations . . . 26

1 Introduction

1.1 The Field of Complex Numbers

Definition 1.1.1 The ***system of complex numbers***, denoted $\mathbb{C}$, is the set $\mathbb{R}^2$ together with the usual rules of ***vector addition*** and ***scalar multiplication*** by a real number α,

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

$$\alpha(x, y) = (\alpha x, \alpha y)$$

and with the operation of ***complex multiplication***, defined by

$$(x_1, y_1)(x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + y_1x_2)$$

Additionally, we define $i := (0, 1)$. For $a, b \in \mathbb{R}$ and $(a, b) \in \mathbb{C}$, the notation $a + bi$ is commonly used, and for $z \in \mathbb{C}$, $z = a + bi = a(1, 0) + b(0, 1)$.

Theorem 1.1.2 The complex numbers $\mathbb{C}$ form a ***field*** satisfying all the field axioms: For all $z, w, s \in \mathbb{C}$,

(i) $z + w = w + z$

(ii) $z+(w+s) = (z+w)+s$

(iii) $z + 0 = z$

(iv) $z + (-z) = 0$

(v) $zw = wz$

(vi) $(zw)s = z(ws)$

(vii) $1z = z$

(viii) $z(z^{-1}) = 1$ for $z \neq 0$

(ix) $z(w + s) = zw + zs$

It is incorrect to extend the domain of the positive square root, $\sqrt{\cdot}$, to the real numbers and write $i = \sqrt{-1}$ assuming all previous properties. Doing so would lead to the contradiction

$$1 = \sqrt{1} = \sqrt{(-1)(-1)} = \sqrt{-1} \cdot \sqrt{-1} = i \cdot i = -1$$

Proposition 1.1.3 For all $z \in \mathbb{C}$ and $z \neq 0$, there exists a unique multiplicative inverse $z^{-1} \in \mathbb{C}$ such that $zz^{-1} = z^{-1}z = 1$.

Proof. Let $z \in \mathbb{C}, z \neq 0$ and $z = a + bi$. Define $z^{-1} = \frac{a}{a^2+b^2} - i\frac{b}{a^2+b^2}$.

$$zz^{-1} = (a, b)(\frac{a}{a^2 + b^2}, -\frac{b}{a^2 + b^2}) = (\frac{a^2 + b^2}{a^2 + b^2}, \frac{-ba + ba}{a^2 + b^2}) = (1, 0)$$

To prove uniqueness, suppose $\exists z_2$ s.t. $zz_2 = 1$.

$$z^{-1} = z^{-1} \cdot 1 = z^{-1} \cdot (z \cdot z_2) = (z^{-1} \cdot z) \cdot z_2 = 1 \cdot z_2 = z_2$$

■

Proposition 1.1.4 For all $z \in \mathbb{C}$, there exists $w \in \mathbb{C}$ such that $w^2 = z$ (Notice that $-w$ also satisfies this equation).

Proof. Let $z \in \mathbb{C}$, $z = a+bi$ $(a, b \in \mathbb{R})$. Let $w = x+iy$ $(x, y \in \mathbb{R})$ s.t. $w^2 = z$ so $w = (a^2 - b^2, 2ab)$. Hence we solve for x, y where $x = a^2 - b^2, y = 2ab$.
Since $(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2 = a^2 + b^2$, $x^2 + y^2 = \sqrt{a^2 + b^2}$. Hence $x^2 = \frac{a+\sqrt{a^2+b^2}}{2}$ and $y^2 = \frac{-a+\sqrt{a^2+b^2}}{2}$. Then we have $x = \pm\sqrt{\frac{a+\sqrt{a^2+b^2}}{2}}$ and $y = \pm\sqrt{\frac{-a+\sqrt{a^2+b^2}}{2}}$, with the sign of x and y depending on the sign of b. ■

1.2 Properties

A complex number may be thought of geometrically as a (two-dimensional) vector, where the horizontal axis is called the ***real axis*** and the vertical axis is called the ***imaginary axis***.

For all $z \in \mathbb{C}$ where $z = (a, b) = a + ib$,

The ***norm*** or ***modulus*** is denoted $|z| = \sqrt{a^2 + b^2}$

The ***argument*** is the angle $\theta = \arg z$ between the vector and the positive direction of the real axis, and $\tan\theta = b/a$. When the strip (interval) is chosen to be $[0, 2\pi)$, θ is the principal argument.

Since $a = r\cos\theta$ and $b = r\sin\theta$, we have

$$a + bi = r\cos\theta + (r\sin\theta)i = r(\cos\theta + i\sin\theta)$$

which is the ***polar representation*** of a complex number.

Proposition 1.2.1 For any $z_1, z_2 \in \mathbb{C}$,

$$|z_1 z_2| = |z_1| \cdot |z_2| \quad \text{and} \quad \arg(z_1 z_2) = \arg z_1 + \arg z_2 \pmod{2\pi}$$

Proof. Let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. Then

$$\begin{aligned} z_1 z_2 &= r_1 r_2[(\cos\theta_1 \cdot \cos\theta_2 - \sin\theta_1 \cdot \sin\theta_2) + i(\cos\theta_1 \cdot \sin\theta_2 + \cos\theta_2 \cdot \sin\theta_1)] \\ &= r_1 r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)] \end{aligned}$$

■

Multiplication of complex numbers can be analyzed in another useful way. Let $z \in \mathbb{C}$ and define $\psi_z : \mathbb{C} \to \mathbb{C}$ by $\psi_z(w) = wz$. By Proposition 1.2.1, the effect of this map is to rotate a complex number through an angle to $\arg z$ in the counterclockwise direction and to stretch its length by the factor $|z|$. In particular, ψ_i rotates complex numbers by $\pi/2$ in the counterclockwise direction. The map ψ_z is a ***linear transformation*** on the plane,

$$\psi_z(\lambda w_1 + \mu w_2) = \lambda\psi_z(w_1) + \mu\psi_z(w_2)$$

where $\lambda, \mu \in \mathbb{R}$ and $w_1, w_2 \in \mathbb{C}$.

If $z = a + bi = (a, b)$, then the matrix of ψ_z is

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} \quad \text{since} \quad \begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax - by \\ ay + bx \end{pmatrix}$$

Proposition 1.2.2 A matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ represents multiplication if and only if $a = d$ and $b = -c$. The complex number in question is $a + ci$.

Proof. Suppose $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = z \cdot (x + yi)$ where $z = \alpha + i\beta \in \mathbb{C}$.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix} = (ax+by)+i(cx+dy) = (\alpha x-\beta y)+i(\alpha y+\beta x) = (\alpha+\beta i)(x+yi) = z\cdot(x+yi)$$

Hence $ax+by = \alpha x-\beta y$ and $cx+dy = \alpha y+\beta x$. Since they hold for any (x, y). Set $x = 0, y = 1$, then $b = -\beta$, $\alpha = d$. Set $x = 1, y = 0$, then $a = \alpha$, $\beta = c$. Therefore, $a = d$, $b = -c$. The matrix representing the multiplication is $\begin{pmatrix} a & -c \\ c & a \end{pmatrix}$.

Now consider multiplying by $a + ci$.

Let $z = \alpha + \beta i$, then $z \cdot a + ci = (\alpha + \beta i) \cdot (a + ci) = (a\alpha - \beta c) + i(\alpha c + \alpha\beta)$.

$$\begin{pmatrix} a & -c \\ c & a \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = (a\alpha - \beta c) + i(\alpha c + a\beta)$$

Therefore $\begin{pmatrix} a & -c \\ c & a \end{pmatrix}$ represents multiplication by $a + ci$. ■

Proposition 1.2.3 (De Moivre's Formula) If $z = r(\cos\theta + i\sin\theta)$ and $n \in \mathbb{Z}^+$, then

$$z^n = r^n(\cos n\theta + i\sin n\theta)$$

Proof. Prove by induction:
I.B.: When $n = 1$, $z = r(\cos\theta + i\sin\theta)$
I.H.: $z^k = r^k(\cos k\theta + i\sin k\theta)$ for some positive integer k.
We want to prove that $z^{k+1} = r^{k+1}(\cos((k+1)\theta) + i\sin((k+1)\theta))$

$$\begin{aligned}\text{I.S.: } z^k z &= r^k r(\cos k\theta + i\sin k\theta)(\cos\theta + i\sin\theta)\\ &= r^{k+1}((\cos k\theta\cos\theta - \sin k\theta\sin\theta) + i(\cos k\theta\sin\theta + \sin k\theta\cos\theta))\\ &= r^{k+1}(\cos((k+1)\theta) + i\sin((k+1)\theta))\end{aligned}$$

■

We can now investigate the nth root of some complex number.

Let $z, w \in \mathbb{R}$ and that $w = r(\cos\theta + i\sin n\theta)$ and $z = \rho(\cos\psi + i\sin\psi)$. We would like to solve for the equation $z^n = w$. Then de Moivre's formula gives $z^n = \rho^n(\cos n\psi + i\sin n\psi)$. It follows that $\rho = \sqrt[n]{r}$ and $n\psi = \theta + k(2\pi)$ where k is some integer. Thus we have the following corollary:

Corollary 1.2.4 Let $w \in \mathbb{C}, w \neq 0$, and w has polar representation $w = r(\cos\theta + i\sin\theta)$. Then the nth roots of w are given by the n complex numbers

$$z_k = \sqrt[n]{r}\left[\cos\left(\frac{\theta}{n} + \frac{2\pi k}{n}\right) + i\left(\frac{\theta}{n} + \frac{2\pi k}{n}\right)\right] \quad k = 0, 1, ..., n-1$$

In particular, for $w \neq 0$, $z_1, ...z_n$ are all of the n distinct solutions.

To prove that the solutions are distinct, we need the following lemma:

Lemma 1.2.5 For all $a, b \in \mathbb{R}$, $\cos a = \cos b$ and $\sin a = \sin b$ if and only if $a = b + 2\pi k, k \in \mathbb{Z}$

Proof. The $\Leftarrow$ direction is trivial.

$\Rightarrow$: Since $\cos a = \cos b$, $b = a + 2\pi k$ or $b = -a + 2\pi k$. Since $\sin a = \sin b$, $b = a + 2\pi k$ or $b = \pi - a + 2\pi k \quad (k \in \mathbb{Z})$. Suppose for contradiction $b \neq a + 2\pi k$, then $b = -a + 2\pi k = \pi - a + 2\pi k = -a + (2k+1)\pi$, which is a contradiction. ■

Returning to the nth roots problem, for $w \neq 0, r \neq 0$, by lemma 1.2.4 (Angles are unique from $k = 1$ up to n), $z_1, ..., z_n$ are distinct. Also, for $\phi_1 = \cos\left(\frac{\theta}{n} + \frac{2\pi k_1}{n}\right)$ and $\phi_2 = \cos\left(\frac{\theta}{n} + \frac{2\pi k_2}{n}\right)$, we can prove that $\phi_1 = \phi_2 + 2\pi m$ for some $m \in \mathbb{Z}$ if and only if $k_1 = k_2 \bmod n$ (i.e. for $i \in \mathbb{Z}\backslash\{1, ..., n\}$, z_i cannot be a distinct solution).

Proof. Assume $\phi_1 = \phi_2 + 2\pi m$. Thus $\cos\left(\frac{\theta}{n} + \frac{2\pi k_1}{n}\right) = \cos\left(\frac{\theta}{n} + \frac{2\pi k_2}{n}\right) + 2\pi m$. So we have $\frac{2\pi}{n}k_1 = \frac{2\pi}{n}k_2 + 2\pi m \Rightarrow k_1 = k_2 + nm \Rightarrow k_1 = k_2 \bmod n$. ■

There are exactly n nth roots of a (nonzero) complex number, which lie on a circle of radius $|z|$ centered at the origin, equally spaced by angles of $2\pi/n$.

Definition 1.2.6 Let $z = a+ib$. Then the ***complex conjugate*** of z is $\bar{z} = a-ib$. Complex conjugation can be pictured geometrically as reflection in the real axis.

Definition 1.2.7 Let $z = a + ib$. Then $x := \mathrm{Re}(z)$ and $y := \mathrm{Im}(z)$.

Proposition 1.2.8 For all $z, w \in \mathbb{C}$. The following properties hold:

(i) $\overline{(z+w)} = \overline{z} + \overline{w}$

(ii) $\overline{zw} = \overline{z} \cdot \overline{w}$

(iii) $\overline{z/w} = \overline{z}/\overline{w}$ for $w \neq 0$

(iv) $z\overline{z} = |z|^2$ and hence if $z \neq 0$, we have $z^{-1} = \bar{z}/|z|^2$

(v) $\overline{-z} = -\overline{z}$

(vi) $z = \overline{z}$ iff $z \in \mathbb{R}$

(vii) $\mathrm{Re}\ z = (z+\overline{z})/2$ and $\mathrm{Im}\ z = (z-\overline{z})/2i$

(viii) $\overline{\overline{z}} = z$

(ix) $z\overline{w} = \overline{w\overline{z}}$

Proof. Fix $z, w \in \mathbb{C}$ and let $z = a + bi, w = x + yi$, where $a, b, x, y \in \mathbb{R}$.

(i) $\overline{(z+w)} = (a+x) - (b+y)i = (a-bi) + (x-yi) = \overline{z} + \overline{w}$

(ii) $\overline{zw} = \overline{(ax-by) + i(ay+bx)} = (a-bi)(x-yi) = \overline{zw}$

(iii) By (ii), $\overline{w}\overline{(z/w)} = \overline{w(z/w)} = \overline{z}$. Hence, $\overline{(z/w)} = \overline{z}/\overline{w}$

(iv) $z\overline{z} = (a+bi)(a-bi) = a^2 + b^2 = |z|^2$

(v) $\overline{-z} = -a + bi = -(a-bi) = -\overline{z}$

(vi) If $z \in \mathbb{R}, b = 0$ thus $z = a = \overline{z}$. If $z = \overline{z}$ then $a+bi = a-bi$ thus $b = -b$ imples $b = 0$. Hence $z = a$, which is a real number.

(vii) $(z+\overline{z})/2 = (a+bi+a-bi)/2 = 2a/2 = a = \mathrm{Re}\ z$. $(z-\overline{z})/2i = (a+bi-a+bi)/2i = 2bi/2i = b = \mathrm{Im}\ z$

(vii) $\overline{\overline{z}} = \overline{a-bi} = a + bi = z$

(ix) $z\overline{w} = (a+bi)(x-yi) = ax+by-(ay-bx)i = \overline{(ax+by)+(ay-bx)i} = \overline{(a-bi)(x+yi)} = \overline{w\overline{z}}$

■

Proposition 1.2.9 For all $z, w \in \mathbb{C}$. The following properties hold:

(i) $|zw| = |z| \cdot |w|$

(ii) If $w \neq 0$, then $|z/w| = |z|/|w|$

(iii) $|\mathrm{Re}\ z| \leq |z|$ and $|\mathrm{Im}\ z| \leq |z|$

(iv) $|\overline{z}| = |z|$

(v) $|z+w| \leq |z| + |w|$ (triangle inequality)

(vi) $|z-w| \geq ||z| - |w||$

(vii) $|z_1w_1 + ... + z_nw_n| \leq \sqrt{|z_1|^2 + ... + |z_n|^2}\sqrt{|w_1|^2 + ... + |w_n|^2}$ (Cauchy-Scharwz inequality)

Proof. Fix $z, w \in \mathbb{C}$ and let $z = a + bi, w = x + yi$, where $a, b, x, y \in \mathbb{R}$.

(i) By Proposition 1.2.1.

(ii) By (i), $|w||z/w| = |w \cdot (z/w)| = |z|$. Hence, $|z/w|/|z|/|w|$.

(iii) Since $b^2 \geq 0, -\sqrt{a^2+b^2} \leq a \leq \sqrt{a^2+b^2}$. Since $a^2 \geq 0, -\sqrt{a^2+b^2} \leq b \leq \sqrt{a^2+b^2}$

(iv) $|\overline{z}| = \sqrt{a^2+(-b)^2} = \sqrt{a^2+b^2} = |z|$

(v) By Proposition 1.2.5(iv),

$$\begin{aligned}
|z+w|^2 &= (z+w)\overline{(z+w)} \\
&= (z+w)(\overline{z}+\overline{w}) \qquad \text{by Proposition 1.2.5(i)} \\
&= z\overline{z} + w\overline{w} + w\overline{z} + z\overline{w} \\
&= z\overline{z} + w\overline{w} + w\overline{z} + \overline{w\overline{z}} \qquad \text{by Proposition 1.2.5 (ix)} \\
&= |z|^2 + |w|^2 + 2\,\mathrm{Re}\ w\overline{z} \qquad \text{by Proposition 1.2.5 (iv) and (vii)} \\
&\leq |z|^2 + |w|^2 + 2|zw| \qquad \text{by (iii)} \\
&= |z|^2 + |w|^2 + 2|z||w| \qquad \text{by (i)} \\
&= (|z| + |w|)^2
\end{aligned}$$

Hence, $z + w \leq |z| + |w|$.

(vi) |z| = |w + (z - w)|≤|w| + |z - w| by (v)

Thus $|z|-|w| \leq |z-w|$. Similarly interchanging z and w, $|w|-|z| \leq |z-w|$ thus $-(|z|+|w|) \leq |z-w|$.

(vii) If $w_k = 0$, then by (v), (vii) can be proven by induction. Assume not all the $w_k = 0$. Let

$$v = \sum_{k=1}^{n} |z_k|^2, \quad t = \sum_{k=1}^{n} |w_k|^2, \quad s = \sum_{k=1}^{n} z_k w_k, \quad c = s/t$$

$$\begin{aligned}
\sum_{k=1}^{n} |z_k - c\overline{w_k}|^2 &= \sum_{k=1}^{n} (z_k - c\overline{w_k})\overline{(z_k - c\overline{w_k})} = \sum_{k=1}^{n} (z_k - c\overline{w_k})(\overline{z_k} - \overline{c\overline{w_k}} \\
&= |z_k|^2 - \overline{c}\sum_{k=1}^{n} z_k w_k - c\sum_{k=1}^{n} \overline{z_k w_k} + |c|^2 |w_k|^2 \\
&= v + |c|^2 t - \overline{c}\sum_{k=1}^{n} z_k w_k - c\sum_{k=1}^{n} \overline{z_k w_k} = v + |c|^2 t - \overline{c}\sum_{k=1}^{n} z_k w_k - \overline{\overline{c}\sum_{k=1}^{n} z_k w_k} \\
&= v + |c|^2 t - 2\,\mathrm{Re}\ \overline{c}s = v + \frac{|s|^2}{t} - 2\,\mathrm{Re}\ \frac{\overline{s}s}{t} \\
&= v + \frac{|s|^2}{t} - 2\frac{|s|^2}{t} = v - \frac{|s|^2}{t} \geq 0
\end{aligned}$$

Hence, $|s|^2 \leq vt$.

■

1.3 Elementary Functions

Definition 1.3.1 For all $z \in \mathbb{C}$, $f : \mathbb{C} \to \mathbb{C}$ is periodic with period $w \in \mathbb{C}\backslash\{0\}$ if

$$f(z+w) = f(z)$$

The ***complex exponential*** $\exp_{\mathbb{C}}$ is a periodic function and is given by the following definition:

Definition 1.3.2 For all $z \in \mathbb{C}$ and $z = x + yi$ $(x, y \in \mathbb{R})$,

$$\exp_{\mathbb{C}}(z) := e^x(\cos y + i \sin y)$$

Remark 1.3.3 Note that we have $e^{iy} = \cos y + i \sin y$ and $e^{-iy} = \cos y - i \sin y$ from the definition.

Hence geometrically, $e^{iy} = (\cos y, \sin y)$ moves in the counterclockwise direction on the unit circle with argument y which goes from 0 to 2π. In exponential form, the polar representation of a complex number is

$$z = |z|e^{i(\arg z)}$$

which is abbreviated to $z = re^{ie}$.

Proposition 1.3.4 For all $z, w \in \mathbb{C}, z = a + ib$ where $a, b \in \mathbb{R}$,

(i) $e^{z+w} = e^z e^w$

(ii) Let $k \in \mathbb{Z}^+$. $e^{kz} = \underbrace{e^z \cdot \ldots \cdot e^z}_{k \text{ terms}}$

(iii) e^z is never zero.

(iv) $|e^{a+bi}| = e^a$; $|e^z| = e^{\mathrm{Re}(z)}$

(v) $e^{\pi i/2} = i, e^{\pi i} = -1, e^{3\pi i/2} = -i, e^{2\pi i} = 1$

(vi) e^z is periodic with period w if and only if $w = 2k\pi i$, for some $k \in \mathbb{Z}\backslash\{0\}$.

(vii) $e^z = 1$ if and only if $z = 2k\pi i$ for some $k \in \mathbb{Z}$

Proof. Fix $z, w \in \mathbb{C}$ and let $z = a + bi, w = x + yi$, where $a, b, x, y \in \mathbb{R}$.

(i) $e^z e^w = e^a(\cos b + i \sin b) \cdot e^x(\cos y + i \sin y) = e^{a+x}(\cos(b+y) + i \sin(b+y)) = e^{(a+x)+i(x+y)} = e^{z+w}$

(ii) Prove by induction:

I.B.: $k = 1$. $e^{kz} = e^{1 \cdot z} = e^z = \underbrace{e^z}_{1 \text{ term}}$

I.H.: Assume that $e^{kz} = \underbrace{e^z \cdot \ldots \cdot e^z}_{k \text{ terms}}$ holds for some $k \in \mathbb{Z}^+$. We want to prove that $e^{(k+1)z} = \underbrace{e^z \cdot \ldots \cdot e^z}_{k+1 \text{ terms}}$.

I.S.:

$$e^{(k+1)z} = e^{kz+z} = e^{kz}e^z = \underbrace{e^z \cdot \ldots \cdot e^z}_{\text{k terms}} \cdot e^z = \underbrace{e^z \cdot \ldots \cdot e^z}_{\text{k + 1 terms}}$$

(iii) $e^z = e^a(\cos b + i \sin b)$ by definition. Since $\cos b$ and $\sin b$ are not both 0, $e^z = 0$ if and only if $e^a = 0$. However $e^a > 0$ for all $x \in \mathbb{R}$. Therefore $e^z \neq 0$.

(iv) $|e^{x+yi}| = |e^x(\cos y + i \sin y)| = \sqrt{(e^x \cos y)^2 + (e^x i \sin y)^2} = \sqrt{(e^x)^2(\cos^2 y + \sin^2 y)} = \sqrt{(e^x)^2} = |e^x| = e^x$

(v) This follows from the definition of $\exp_{\mathbb{C}}$.

(vi) Since $\exp_{\mathbb{C}}$ is periodic, $e^{z+w} = e^z \iff e^z e^w = e^z \iff e^w = 1 (e^z \neq 0) \iff w = 2\pi k i$ where $k \in \mathbb{Z}$

(vii) $e^z = e^a(\cos b + i \sin b)$ by definition.

$$\begin{aligned} e^z = 1 &\iff e^a \cos b = 1 \text{ and } e^a \sin b = 0 \\ &\iff b = 2\pi k \text{ and } a = 0 \text{ where } k \in \mathbb{Z} \\ &\iff z = 2\pi k \text{ where } k \in \mathbb{Z} \end{aligned}$$

■

From *Remark 1.3.3*, we can define the ***complex sine*** and ***complex cosine*** using the complex exponential.

Definition 1.3.5 For all $z \in \mathbb{C}$,

$$\sin_{\mathbb{C}} z := \frac{e^{iz} - e^{-iz}}{2i} \qquad \text{and} \qquad \cos_{\mathbb{C}} z := \frac{e^{iz} + e^{-iz}}{2}$$

We would have come up with this definition if we could use the formulas for Taylor series that we know for real numbers with complex numbers, then evaluate the right hand side and obtain the Taylor series for cosine and sine, though we will leave the discussion of series and convergence to a later section.

Proposition 1.3.6 For all $z, w \in \mathbb{C}$,

(i)
$$\sin^2 z + \cos^2 z = 1$$

(ii)
$$\sin(z + w) = \sin z \cdot \cos w + \cos z \cdot \sin w$$
$$\cos(z + w) = \cos z \cdot \cos w - \sin z \cdot \sin w$$

(iii)
$$\cos(x + iy) = \frac{\cos(x)(e^y + e^{-y})}{2} + i\frac{\sin(x)(-e^y + e^{-y})}{2}$$
$$\sin(x + iy) = \frac{\sin(x)(e^y + e^{-y})}{2} + i\frac{\cos(x)(e^y - e^{-y})}{2}$$

Proof. The five equations follow directly from definitions of complex cosine/sine and the complex exponential. ■

Note that the real cosine and sine are bounded by 1. However, boundedness does not carry onto the complex cosine and sine, as we can see from Proposition 1.3.6 (iii) that e^y can go to infinity.

Lemma 1.3.7 For all $h \in \mathbb{R}$, there exists $z \in \mathbb{C}$ such that $|\cos z| > h, |\sin z| > h$

Proof. Fix $r \in \mathbb{R}$. Choose $y = \max\{\log(2h+1), 1\} + 1$

$$\begin{aligned} y &> \log(2h+1) \\ e^y &> 2h+1 \\ e^y &> 2h + e^{-y} \\ e^{2y} &> e^y 2h + 1 \\ \frac{e^y - e^{-y}}{2} &> h \end{aligned}$$

Additionally, $\frac{e^y+e^{-y}}{2} > h$. Choose $z = (0, y)$. Then $\mathrm{Re}(\cos z) > h$ and $\mathrm{Im}(\sin z) > h$. By Proposition 1.2.8, $|\cos z| \geq |\mathrm{Re}(\cos z)| > h$ and $|\sin z| \geq |\mathrm{Im}(\sin z)| > h$. Further, choose $z_n = 0 + in$ for $n \in \mathbb{N}$ then $|\cos z| \to \infty$ as $n \to \infty$ and hence it is not only not bounded by 1 but the modulus diverges as you travel along the imaginary axis. ■

However, we can see from Proposition 1.3.6 (i) that the sum of squares of the complex sine and cosine is still equal to 1.

Now let's take a look at the ***complex logarithm***. In the real numbers, we can view the logarithm as the inverse of the exponential. When we allow z to range over $\mathbb{C}$, the exponential cannot have a unique inverse because it is periodic. However, restrictions of the complex exponential do have inverses.

Proposition 1.3.8 Let A_{y0} denote the set of complex numbers $z = x + yi$ $(x, y \in \mathbb{R})$ such that

$$A_{y0} = \{x + iy \mid x \in \mathbb{R} \quad \text{and} \quad y_0 \leq y < y_0 + 2\pi\}$$

Then e^z maps A_{y0} bijectively to $\mathbb{C}\backslash\{0\}$.

Proof. From Proposition 1.3.4 (ii), $\exp_{\mathbb{C}}(A_{y0}) \subseteq \mathbb{C}\backslash\{0\}$. We will show that the map $\exp : A_{y0} \to \mathbb{C}\backslash\{0\}$ is both injective and surjective.

First we show that the map is injective.

Fix $z_1, z_2 \in A_{y0}$. Suppose z_1 and z_2 map to the same point, thus $\exp(z_1) = \exp(z_2)$. Hence $\exp(z_1 - z_2) = 1 \Rightarrow z_1 - z_2 = 2\pi ki$ for some $k \in \mathbb{Z}$ by Proposition 1.3.4 (v) and (vi). Since $-2\pi < \mathrm{Im}(z_1) - \mathrm{Im}(z_2) < 2\pi, k = 0$ hence $z_1 = z_2$.

Next we show that the map is surjective.

Fix $z \in \mathbb{C}\backslash\{0\}$. By the polar representation of z, there exists unique $r > 0$ and $\theta \in [0, 2\pi)$ such that $z = r(\cos\theta + i\sin\theta)$. Let $x_0 = \log(r)$. By Lemma 1.2.4, there exists y_0 such that $\cos y_0 = \cos\theta$ and $\sin y_0 = \sin\theta$. Thus there is $x_0 + iy_0 \in A_{y0}$ and $\exp(x_0 + iy_0) = z$. ■

Definition 1.3.9 For $A_y = \{z \in \mathbb{C} : y \leq \mathrm{Im}(z) < y + 2\pi\}$, the branch of the complex logarithm lying in the strip A_y is defined as the function $\log_{A_y} : \mathbb{C}\backslash\{0\} \to A_y$,

$$z \mapsto \exp|_{A_y}^{-1}(z)$$

An alternative definition $\log_{A_y} : \mathbb{C}\backslash\{0\} \to A_y$,

$$z \mapsto \log|z| + i\arg_{A_y}(z)$$

The branch of the complex logarithm lying in the strip $A_0 = \{z \in \mathbb{C} : 0 \leq \mathrm{Im}(z) < 2\pi\}$ is the ***principal branch***.

The function $\log z$ is well defined only when a specific branch is chosen.

Proposition 1.3.10 Let $A_1 = \{z \in \mathbb{C} : y_1 \leq \mathrm{Im}(z) < y_1 + 2\pi\}$ and $A_2 = \{z \in \mathbb{C} : y_2 \leq \mathrm{Im}(z) < y_2 + 2\pi\}$. Then for all $z \in \mathbb{C}, \log_{A_1}(z) - \log_{A_2}(z) = 2\pi ni$ for some $n \in \mathbb{Z}$.

Proof. Fix $z \in \mathbb{C}, y_1, y_2 \in \mathbb{R}$. By definition, $\log_{A_1}(z) = \exp|_{A_1}^{-1}(z)$ and $\log_{A_2}(z) = \exp|_{A_2}^{-1}(z)$. Take exponential on both sides, $e^{\log_{A_1}(z)} = z$ and $e^{\log_{A_2}(z)} = z$. Therefore $e^{\log_{A_1}(z)} = e^{\log_{A_2}(z)} \Rightarrow e^{\log_{A_1}(z) - \log_{A_2}(z)} = 1$. By Proposition 1.3.4 (vi), $\log_{A_1}(z) - \log_{A_2}(z) = 2\pi ni$ for some $n \in \mathbb{Z}$. ■

Proposition 1.3.11 Let $A_y = \{z \in \mathbb{C} : y \leq \mathrm{Im}(z) < y + 2\pi\}$.

(i) The relation $\exp(\log_{A_y}(z)) = z$ holds for all $z \in \mathbb{C}$.

(ii) The relation $\log_{A_y}(\exp(z)) = z$ holds if and only if $z \in A_y$.

Proof. Fix a branch A_y. Fix $z \in \mathbb{C}$

(i) $\exp(\log_{A_y}(z)) = \exp(\exp|_{A_y}^{-1}(z)) = \exp|_{A_y}(\exp|_{A_y}^{-1}(z)) = z$

(ii) Suppose $\log_{A_y}(\exp(z)) = z$. Since the range of $\log_{A_y}$ is A_y, $z \in A_y$.
Suppose $z \in A_y$, $\log_{A_y}(\exp(z)) = \exp|_{A_y}^{-1}(\exp|_{A_y}(z)) = z$

■

Also, $\exp(\log_{A_y}(z)) = z$ holds for *any* branch A_y.

Proposition 1.3.12 Let $A_y = \{z \in \mathbb{C} : y \leq \mathrm{Im}(z) < y + 2\pi\}$. For all $z_1, z_2 \in \mathbb{C}\backslash\{0\}$, $\log_{A_y}(z_1 z_2) = \log_{A_y}(z_1) + \log_{A_y}(z_2) + 2\pi ni$ for some $n \in \mathbb{Z}$.

Proof. Fix a branch A_y. By definition, $\log_{A_y}(z_1 z_2) = \log|z_1 z_2| + i\arg_{A_y}(z_1 z_2)$. Since $\log|z_1 z_2| = \log|z_1||z_2| = \log|z_1| + \log|z_2|$ by properties of the real logarithm, and $\arg_{A_y}(z_1 z_2) = \arg_{A_y}(z_1) + \arg_{A_y}(z_2) + 2\pi ni$ for some $n \in \mathbb{Z}$, $\log_{A_y}(z_1 z_2) = (\log|z_1| + i\arg_{A_y}(z_1)) + (\log|z_2| + i\arg_{A_y}(z_2)) + 2\pi ni$ for some $n \in \mathbb{Z}$. ■

With the complex logarithm, we can now define the ***complex powers***, the expression a^b where $a, b \in \mathbb{C}$ and $a \neq 0$.

Definition 1.3.13 Let $a, b \in \mathbb{C}$ with $a \neq 0$. Given a branch of the complex logarithm, $\log_A$, the complex power specific to the branch of logarithm is defined as

$$a^b = e^{b \log_A a}$$

Note that a^b is multiple-valued. For example, if $b = 1/q, q \in \mathbb{Z}$, then a^b is one of the qth roots of a and has q distinct values.

Theorem 1.3.14 Let $a, b, \in \mathbb{C}, a \neq 0$. Then there are three cases for a^b

1. If $b \in \mathbb{Z}$, a^b has a unique value.
2. If $b \in \mathbb{Q}, b \neq 0$ ($b = p/q, q \in \mathbb{N}$, p/q in irreducible form), a^b has exactly q unique values depending on the choice of the branch.
3. If $b \notin \mathbb{Q}$, a^b has infinitely many unique values depending on the choice of the branch. Any two values differ by a factor of $e^{2\pi nbi}$ for $n \in \mathbb{Z}$.

Proof. Fix $a, b \in \mathbb{C}, a \neq 0$. Fix a branch A of logarithm.

(i) Fix $b \in \mathbb{Z}$. For $b > 0$,

$$a^b = e^{b \log_A(a)} = \underbrace{e^{\log_A(a)} \cdot \ldots \cdot e^{\log_A(a)}}_{b \text{ terms}} = \underbrace{a \cdot \ldots \cdot a}_{b \text{ terms}}$$

For $b = 0$, $a^0 = e^{0 \log_A(a)} = e^0 = 1$.

For $b < 0$, $a^0 = e^{b \log_A(a)} = a^0 = (e^{-b \log_A(a)})^{-1} = (\underbrace{a \cdot \ldots \cdot a}_{-b \text{ terms}})^{-1}$.

There is a unique value of a^b for each of the cases above.

(ii) Fix $b \in \mathbb{Q}, b \neq 0$, so $b = p/q, q \in \mathbb{Z} \backslash \{0\}$ in irreducible form.

Let A_0 be the principal branch for the logarithm of a and some other branch be $A = \{z \in \mathbb{C} : y \leq \operatorname{Im}(z) < y + 2\pi\}$, then $\log_A(a) = \log_{A_0}(a) + 2N\pi i$ for some $N \in \mathbb{Z}$.

$$\begin{aligned} a^b &= \exp(b \log_A(a)) \\ &= \exp(\frac{p}{q} \log_A(a)) \\ &= \exp(p(\frac{1}{q}(\log_{A_0}(a) + 2N\pi i))) \\ &= \underbrace{\exp(\frac{\log_{A_0}(a)}{q} + \frac{2\pi N}{q} i) \cdot \ldots \cdot \exp(\frac{\log_{A_0}(a)}{q} + \frac{2\pi N}{q} i)}_{p \text{ terms}} \end{aligned}$$

We want to show that there are q distinct values of N. Fix two other branches $A_1 = \{z \in \mathbb{C} : y_1 \leq \operatorname{Im}(z) < y_1 + 2\pi\}$ such that $\log_{A_1}(a) = \log_{A_0}(a) + 2n\pi i$ for some $n \in \mathbb{Z}$,

$A_2 = \{z \in \mathbb{C} : y_2 \leq \mathrm{Im}(z) < y_2 + 2\pi\}$ such that $\log_{A_2}(a) = \log_{A_0}(a) + 2m\pi i$ for some $m \in \mathbb{Z}$.

$$\begin{aligned}\frac{\exp(b\log_{A_1}(a))}{\exp(b\log_{A_2}(a))} &= \underbrace{\exp(2\pi\frac{m-n}{q}i)\cdot\ldots\cdot\exp(2\pi\frac{m-n}{q}i)}_{p \text{ terms}} \\ &= \exp(2\pi\frac{p}{q}(m-n)i)\end{aligned}$$

We also have

$$\begin{aligned}\exp(b\log_{A_1}(a)) = \exp(b\log_{A_2}(a)) &\iff \frac{\exp(b\log_{A_1}(a))}{\exp(b\log_{A_2}(a))} \\ &\iff \frac{p}{q}(m-n) \in \mathbb{Z} \\ &\iff (m-n) = kq \text{ for some } k \in \mathbb{Z}\end{aligned}$$

Thus N correspond to $(m-n)$. For choices of A such that $N = 0, ..., q-1$, each N gives a distinct $\exp(b\log_A(a))$. For choices of A such that N takes on integers other than $0, ..., q-1$, there exists $N' \in \{0, ..., q-1\}$ such that $N - N' = tq$ for some $t \in \mathbb{Z}$ and hence do not give a distinct $\exp(b\log_A(a))$. Therefore, a^b takes on q distinct values.

(iii) Fix $b \notin \mathbb{Q}$. Proof to be completed

■

2 Basic Topology

2.1 Open and closed sets

Definition 2.1.1 Let $z_0 \in \mathbb{C}, r \in \mathbb{R}_{>0}$. Let $A \subseteq \mathbb{C}$.
The r ***neighbourhood*** of z_0 with radius r is the open ball

$$B_r(z_0) = \{z \in \mathbb{C} : |z - z_0| < r\}$$

z_0 is an interior point of A if there exists r such that $B_r(z_0) \subseteq A$.
The ***deleted*** r ***neighbourhood*** is the set

$$B_r(z_0)\backslash\{z_0\}$$

z_0 is a limit point of A if for all r, $B_r(z_0) \cap A\backslash\{z_0\} \neq \varnothing$.

Definition 2.1.2 A set $A \subseteq \mathbb{C}$ is ***open*** if for all points $z_0 \in A$, there exists $\varepsilon \in \mathbb{R}, \varepsilon > 0$, such that $B_\varepsilon(z_0) \subseteq A$.

Definition 2.1.3 A set $A \subseteq \mathbb{C}$ is ***closed*** if its complement $A^{\complement} := \mathbb{C}\backslash\{A\}$ is open.

Closed sets can also be characterized by convergent sequences.

Proposition 2.1.4 Let $C \subseteq \mathbb{C}$ be a set. C is closed if and only if every sequence $\{z_n\}$ in C, such that $\lim_{n\to\infty} z_n$ exists, has $\lim_{n\to\infty} z_n \in \mathbb{C}$.

Proof. ($\Longrightarrow$):

Let $C \subseteq \mathbb{C}$ be a closed set and suppose for contradiction some sequence $\{z_n\}$ has $\lim_{n\to\infty} z_n = z_0 \in C^{\complement}$. $C^{\complement}$ is open since C is closed. Thus there exists $\varepsilon > 0$ such that $B_\varepsilon(z) \subseteq C^{\complement}$. However, since $\{x_n\} \in C$, there does not exists $N \in \mathbb{N}$ such that for all $n > N, z_n \in B_\varepsilon(z_0)$, which is a contradiction. Therefore, every convergent sequence converges to a point in C.

($\Longleftarrow$):

Let $C \subseteq \mathbb{C}$ be a set and suppose for contradiction it is not closed. $\overline{C} = C \cup \partial C$[1]. Let $C \subseteq \mathbb{C}$ be a set and suppose for contradiction it is not closed. Fix $z' \in \partial C \backslash C$. Consider the sequence $\{z_n\}$ s.t. $z_n \in B_{1/n}(z) \cap C$ for each $n \in \mathbb{N}$. Then $\{z_n\} \in C$ but converges to $z' \in C$, which is a contradiction. ■

Proposition 2.1.5 All of the following are true for sets in $\mathbb{C}$:

(i) The empty set $\varnothing$ is open and closed.

(ii) $\mathbb{C}$ is open and closed.

(iii) A finite or infinite union of open sets is open.

(iv) A finite intersection of open sets is open.

(v) A finite union of closed sets is closed.

(vi) A finite or infinite intersection of closed sets is closed.

Proof. (i) The empty set is vacuously open and closed.

(ii) Fix $z_0 \in \mathbb{C}$ and choose $r = 1$, $D_1(z_0) \subset \mathbb{C}$. Thus $\mathbb{C}$ is open. $\mathbb{C}^C = \emptyset$ which is open. Therefore $\mathbb{C}$ is closed.

(iii) Proof to be completed

(iv) Proof to be completed

(v) Proof to be completed

(vi) Proof to be completed

■

[1] Closure $\overline{C}$ of C is the set C along with its limit points. The topological boundary ∂C of C is the set of boundary points of C. A boundary point of C is is a point $z \in \mathbb{C}$ such that for all $\varepsilon > 0, B_\varepsilon(z) \cap C$ and $B_\varepsilon(z) \cap C^{\complement}$ are both non-empty.

Definition 2.1.6[a] Let $S \subseteq \mathbb{C}$ be a set.

$A \subseteq S$ is open relative to S if and only if $A = S \cap O$ for some open set O.

$B \subseteq S$ is closed relative to S if and only if $B = S \cap C$ for some closed set C.

[a] The precise construction is known as "subspace topology", which we will not go into detail in this set of notes.

Note that if A is relatively open, then $S \backslash A$ is relatively closed.

2.2 Limits and continuity

Definition 2.2.1 Let $f : A \subseteq \mathbb{C} \to \mathbb{C}$. Let $z_0 \in \mathbb{C}$ be a limit point of A. Define a to be the ***limit of*** f ***at*** z_0 if

$$\forall \varepsilon, \exists \delta > 0, \text{s.t. } \forall z \in A, 0 < |z - z_0| < \delta \Rightarrow |f(z) - a| < \varepsilon$$

If the above holds, then write $\lim_{z \to z_0} = a$.

Similar to the real case, this says informally,

"The value f(z) can be made ε-close to a if z is δ-close to z_0"

Note that a lot of definitions, such as the definition of limit above, carries from multivariable calculus, as we can treat the space of complex numbers as an $\mathbb{R}^2$ space. However, as we will see, not all the properties are the same in $\mathbb{C}$ and $\mathbb{R}^2$.

Proposition 2.2.2 Limits are unique if they exist.

Proof. Let $f : A \subseteq \mathbb{C} \to \mathbb{C}$ and fix a limit point z_0 of A. Suppose there exists $a, b \in \mathbb{C}, a \neq b$ such that $\lim_{z \to z_0} = a$ and $\lim_{z \to z_0} = b$. For all $\varepsilon > 0$, there exists $\delta_1, \delta_2 > 0$ such that for all $z \in A$

$$0 < |z - z_0| < \delta_1 \Rightarrow |f(z) - a| < \varepsilon \quad \text{and} \quad 0 < |z - z_0| < \delta_2 \Rightarrow |f(z) - b| < \varepsilon$$

Choose $\delta = \min(\delta_1, \delta_2)$, then $0 < |z - z_0| < \delta \Rightarrow |f(z) - a| < \varepsilon$ and $|f(z) - b| < \varepsilon$.

$$|a - f(z)| + |f(z) - b| = |a - b| < 2\varepsilon$$

Since this holds for arbitrary $\varepsilon > 0$, we must have $a = b$. ■

Proposition 2.2.3 Let $A \subseteq \mathbb{C}$ and let z_0 be a limit point of A. Let $f, g : A \to \mathbb{C}$. All the following hold:

(i) If $\lim_{z\to z_0} f(z)$ and $\lim_{z\to z_0} g(z)$ exist then $\lim_{z\to_0}(f(z)+g(z))$ exist and

$$\lim_{z\to z_0}(f(z)+g(z)) = \lim_{z\to z_0} f(z) + \lim_{z\to z_0} g(z)$$

(ii) If $\lim_{z\to z_0} f(z)$ and $\lim_{z\to z_0} g(z)$ exist then $\lim_{z\to z_0}(f(z)\cdot g(z))$ exist and

$$\lim_{z\to z_0}(f(z)\cdot g(z)) = (\lim_{z\to z_0} f(z))\cdot(\lim_{z\to z_0} g(z))$$

(iii) If $\lim_{z\to z_0} f(z)$ and $\lim_{z\to z_0} g(z)$ exist, and $\lim_{z\to z_0} g(z) \neq 0$ then $\lim_{z\to z_0}(f(z)/g(z))$ exist and

$$\lim_{z\to z_0}(f(z)/g(z)) = (\lim_{z\to z_0} f(z))/(\lim_{z\to z_0} g(z))$$

Proof. Let $f, g : A \subseteq \mathbb{C} \to \mathbb{C}$. Let $\lim_{z\to z_0} f(z) = a$ and $\lim_{z\to z_0} g(z) = b$. For all $\varepsilon > 0$, there exists $\delta_1, \delta_2 > 0$ such that for all $z \in A$,

$$0 < |z - z_0| < \delta_1 \Rightarrow |f(z) - a| < \varepsilon$$

$$0 < |z - z_0| < \delta_2 \Rightarrow |g(z) - b| < \varepsilon$$

(i) Choose $\delta = \min(\delta_1, \delta_2)$. Then

$$|(f(z)+g(z)) - (a+b)| \leq |f(z) - a| + |g(z) - b| < 2\varepsilon$$

by triangle inequality. Since this holds for arbitrary $\varepsilon > 0$, we have $\lim_{z\to z_0}(f(z)+g(z)) = a + b = \lim_{z\to z_0} f(z) + \lim_{z\to z_0} g(z)$

(ii) $f(z)g(z) - ab = (f(z)-a)(g(z)-b) + (f(z)-a)b + (g(z)-b)a$. By the triangle inequality,

$$|f(z)g(z) - ab| \leq |f(z)-a||g(z)-b| + |f(z)-a||b| + |g(z)-b||a|$$

Fix $\varepsilon > 0$. Choose $\delta > 0$ such that for $z \in B_\delta(z_0)\backslash\{z_0\}$, $|f(z)-a| < \varepsilon/3, |g(z)-b| < 1, |f(z)-a| < \varepsilon/(3|b|), |g(z)-b| < \varepsilon/(3|a|)$. (Choose the smallest of the four different δ_i that satisfy these four conditions).

(iii) By (ii), it suffices to show $\lim_{z\to z_0}(1/g)(z) = 1/b$.

$$\begin{aligned}\left|\frac{1}{g(z)} - \frac{1}{b}\right| &= \left|\frac{g(z)-b}{g(z)b}\right| \\ &= \frac{|g(z)-b|}{|g(z)||b|} \\ &\leq ||\end{aligned}$$

Proof to be completed

■

A function is said to be continuous at a point if the limit exists at that point and its agrees with the value of the function at that point.

Definition 2.2.4 Let $A \subseteq \mathbb{C}$ be a set and let $f : A \to \mathbb{C}$ be a function. f is ***continuous at*** $z_0 \in A$ if

$$\forall \varepsilon, \exists \delta > 0, \text{s.t. } \forall z \in A, |z - z_0| < \delta \Rightarrow |f(z) - f(z_0)| < \varepsilon$$

which is that

$$\lim_{z \to z_0} f(z) = f(z_0)$$

f is ***continuous on*** A if f is continuous at each point z_0 in A.

Theorem 2.2.5 For a function $f : A \subseteq \mathbb{C} \to \mathbb{C}$:

(i) If $z_0 \in A$ is a limit point of A (not an isolated point), f is continuous at z_0 if and only if $\lim_{z \to z_0} f(z) = f(z_0)$.

(ii) If $z_0 \in A$ is not a limit point of A (an isolated point), f is automatically continuous at z_0.

Proof. Let $f : A \subseteq \mathbb{C} \to \mathbb{C}$. Fix $z_0 \in A$.

(i) Suppose z_0 is a limit point of A. This is true by definition.

(ii) Suppoe z_0 is a not limit point of A. Then there exists $\delta > 0$ such that $B_\delta(z_0) \cap A = \{x\}$. Choose this δ, thus for all $\varepsilon > 0$, for all $z \in B_\delta(z_0) \cap A$, $|f(z) - f(z_0)| = 0 < \varepsilon$ (there is only one point in this open ball). Thus f is continuous at isolated points.

■

From Proposition 1.4.8, we can deduce that if f and g are continuous on A, then so are $f + g$ and $f \cdot g$, f/g if $g(z_0) \neq 0$. We also have that the composition of continuous functions is continuous.

Theorem 2.2.6 Let $f : A \to B$ where $A \subseteq \mathbb{C}$ and $B \subseteq \mathbb{C}$. Let $g : B \to \mathbb{C}$.

(i) Let a be a limit point of A and let $b \in B$. If $\lim_{z \to a} f(z) = b$ and g is continuous at b then $\lim_{z \to a} g \circ f(z) = g(b)$.

(ii) Let $a \in A$. If f is continuous at a and g is continuous at $f(a)$ the $g \circ f$ is continuous at a.

Proof. Let $f : A \to B$ where $A \subseteq \mathbb{C}$ and $B \subseteq \mathbb{C}$. Let $g : B \to \mathbb{C}$.

(i) Assume $\lim_{z \to z_0} f(z) = b$ and g is continuous at b. Fix $\varepsilon > 0$. There exists $\eta > 0$ such that for all $z_b \in B$,

$$|z_b - b| < \eta \Longrightarrow |g(z_b) - g(b)| < \varepsilon$$

Also, since $\eta > 0$, there exists $\delta > 0$ such that for all $x \in A$,

$$|0 < |z - a| < \delta| \Longrightarrow |f(z) - b| < \eta$$

Fix $x \in A$. Assume $0 < |z - a| < \delta$. By the above with $z_b = f(z)$, $|g(f(z)) - g(b)| < \varepsilon$

(ii) This follows directly from (i).

■

Proposition 2.2.7 Let $f : \mathbb{C} \to \mathbb{C}$ be any function, the following are true for any $A, B \subseteq \mathbb{C}$

(i) $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$

(ii) $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$

(iii) $f^{-1}(A) \subseteq f^{-1}(B)$ if $A \subseteq B$

(iv) $f^{-1}(A \backslash B) = f^{-1}(A) \backslash f^{-1}(B)$

Proof. We will proof (i). The others are proven in a similar way.

Let $z \in f^{-1}(A \cup B) \implies f(z) = A \cup B \implies f(z) \in A$ or $f(z) \in B \implies z \in f^{-1}(A)$ or $z \in f^{-1}(B)$ Hence $z \in f^{-1}(A) \cup f^{-1}(B) \implies f^{-1}(A \cup B) \subseteq f^{-1}(A) \cup f^{-1}(B)$.

Let $z \in f^{-1}(A) \cup f^{-1}(B) \implies z \in f^{-1}(A)$ or $z \in f^{-1}(B) \implies f(z) \in A$ or $f(z) \in B \implies f(z) = A \cup B \implies z \in f^{-1}(A \cup B)$. Hence $f^{-1}(A) \cup f^{-1}(B) \subseteq f^{-1}(A \cup B)$.

Therefore, $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$. ■

Theorem 2.2.8 Let $f : \mathbb{C} \to \mathbb{C}$. The following are equivalent:

(i) f is continuous on C.

(ii) The preimage $f^{-1}(U)$ is open for every open set $U \subseteq \mathbb{C}$.

(iii) The preimage $f^{-1}(V)$ is closed for every closed set $V \subseteq \mathbb{C}$.

Proof. Let $f : \mathbb{C} \to \mathbb{C}$.

(i) $\Rightarrow$ (ii): Assume f is continuous on C. By definition,

$$\forall z_0 \in \mathbb{C}, \forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } f(D_\delta(z)) \subseteq D_\varepsilon(f(z_0))$$

$$\Leftrightarrow \forall z_0 \in \mathbb{C}, \forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } D_\delta(z_0) \subseteq f^{-1}(D_\varepsilon(f(z_0)))$$

Fix an open set $U \subseteq \mathbb{C}$. Let $z_0 \in f^{-1}(U)$ so $f(z_0) \in U$. Since U is open, there exists $\varepsilon > 0$ s.t. $D_\varepsilon(f(z_0)) \subseteq U \Rightarrow f^{-1}(D_\varepsilon(f(z_0))) \subseteq f^{-1}(U)$. By the above definition, there exists $\delta > 0$ such that $D_\delta(a) \subseteq f^{-1}(D_\varepsilon(f(z_0))) \subseteq f^{-1}(U)$. Therefore $f^{-1}(U)$ is open.

(i) $\Rightarrow$ (iii): Fix a closed set $V \subseteq \mathbb{C}$ so that V^C is open. Following from the proof above, $f^{-1}(V^C) = (f^{-1}(V))^C$ is open. Therefore $f^{-1}(V)$ is closed.

(ii) $\Rightarrow$ (i): Fix a open set $U \subseteq \mathbb{C}$. Fix $z_0 \in U$. Proof to be completed

■

Proposition 2.2.9 Let $f : A \subseteq \mathbb{C} \to \mathbb{C}$ and $u = \mathrm{Re}(f), v = \mathrm{Im}(f)$. Let z_0 be a limit point of A. $\lim_{z\to z_0} f(z)$ exists if and only if $\lim_{z\to z_0} u(z)$ and $\lim_{z\to z_0} v(z)$ exists. Then

$$\lim_{z\to z_0} f(z) = \lim_{z\to z_0} u(z) + \lim_{z\to z_0} v(z)i$$

Proof. ($\Longrightarrow$): Let $f : A \subseteq \mathbb{C} \to \mathbb{C}$ and $u = \mathrm{Re}(f), v = \mathrm{Im}(f)$. Suppose $\lim_{z\to z_0} u(z)$ and $\lim_{z\to z_0} u(z)$ exists. Then by Proposition 2.2.3, $\lim_{z\to z_0} f(z)$ exists and $\lim_{z\to z_0} f(z) = \lim_{z\to z_0} u(z) + \lim_{z\to z_0} v(z)i$.

($\Longleftarrow$): Suppose $\lim_{z\to z_0} f(z)$ exists and $\lim_{z\to z_0} f(z) = l$. Then for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $z \in A$,

$$0 < |z - z_0| < \delta \Longrightarrow |f(z) - l| < \varepsilon$$

Claim that $\lim_{z\to z_0} u(z)$ exists and is equal to $\mathrm{Re}(l)$. Fix $\varepsilon > 0, z \in A \cap B_\delta(z_0)\backslash z_0$ and choose the same δ. Then we have $|f(z) - l| < \varepsilon$.

$$\begin{aligned}
|u(z) - \mathrm{Re}(l)| &= \sqrt{(u(z) - \mathrm{Re}(l))^2} \\
&\leq \sqrt{(u(z) - \mathrm{Re}(l))^2 + (v(z) - \mathrm{Im}(l))^2} \\
&= |f(z) - l| \\
&< \varepsilon
\end{aligned}$$

Similarly, $\lim_{z\to z_0} v(z)$ exists and is equal to $\mathrm{Im}(l)$. ■

Definition 2.2.10 Let $A \subseteq \mathbb{C}$ be a set. A function $f : A \to \mathbb{C}$ is ***uniformly continuous*** if

$$\forall \epsilon > 0, \exists \delta > 0, \text{ s.t. } \forall z_1, z_2 \in A, |x_1 - x_2| < \delta \Longrightarrow |f(z_1) - f(z_2)| < \epsilon$$

Uniform continuity is a *global* property of a function which does not depend on any particular point, as compared to continuity that we defined earlier in this section.

2.3 Connected and path-connected sets

Definition 2.3.1 A set $A \subseteq \mathbb{C}$ is ***path-connected*** if for every pair of points $z_0, z_1 \in A$, there exists a continuous function $\gamma : [0, 1] \to \mathbb{C}$ such that $\gamma(0) = z_0$ and $\gamma(1) = z_1$.

The definition of "connected" is more complicated. We first define "not connected" and take "connected" to be the negation.

Definition 2.3.2 A set $C \subset \mathbb{C}$ is ***not connected*** if there exists open sets U and V such that

(i) $C \subset U \cup V$

(ii) $C \cap U \neq \varnothing$ and $C \cap V \neq \varnothing$

(iii) $(C \cap U) \cap (C \cap V) = \varnothing$

If a set is not "not connected", it is ***connected***.

Lemma 2.3.3 A set $S \subseteq \mathbb{C}$ is connected if and only if there does not exist $D \subseteq S$ such that $D \neq \varnothing, D \neq S$, and D is relatively clopen.

Proof. ($\Longrightarrow$) by proving the contrapositive:

Suppose there exists $D \subseteq S$ such that $D \neq \varnothing, D \neq S$, and D is relatively clopen. Since D is relatively closed, $S \backslash D$ is relatively open. Since D is relatively open to S, there exists open set U such that $S \cap U = D$. Since $S \backslash D$ is relatively open to S, there exists open set V such that $S \cap V = S \backslash D$. U and V satisfy the conditions of Definition 2.3.2. Therefore S is not connected.

($\Longleftarrow$) by proving the contrapositive:

Suppose S is not connected. Then there exists open sets U and V that satisfy the conditions of Definition 2.3.2. Let $D = S \cap U, C = S \cap V$. Thus $D \neq \varnothing$. Since $S \subseteq U \cup V, S = D \cup C$. Since $(S \cap U) \cap (S \cap V) = \varnothing, C = S \backslash D$. Since $D \neq \varnothing$ and $C \neq \varnothing, D \neq S$. Since C is relatively open, D is relatively closed. Hence, $D \subseteq S$ satistifies that $D \neq \varnothing$, $D \neq \varnothing, D \neq S$, and D is relatively clopen. ■

Theorem 2.3.4 Every path-connected set is connected.

Proof. Fix $A \subseteq \mathbb{C}$ which is path-connected. Suppose for contradiction that A is not connected. By definition, there exists open sets U and V such that $A \subset U \cup V$, $C \cap U \neq \emptyset$ and $C \cap V \neq \emptyset$ and $(C \cap U) \cap (C \cap V) = \emptyset$.

Let $U' = S \cap U$ and $V' = S \cap V$. Fix $a \in U'$ and $b \in V'$. Suppose there exists some continuity $\gamma : [0,1] \to S$ such that $\gamma(0) = a$ and $\gamma(1) = b$.

Let $A = \{t \in [0,1] : \gamma(t) \in U'\}$. Since A is non-empty ($0 \in A$ since $\gamma(0) = a \in U'$) and bounded above by 1, $\sup A = u \in [0,1]$ exists.

Suppose $\gamma(u) \in U'(\gamma(u) \notin V', u \neq 1)$, then $\gamma(u) \in U$. Since U is open, there exists $\varepsilon > 0$ such that $D_\varepsilon(\gamma(u)) \subseteq U$. By continuity of γ, there exists $\delta > 0$ such that $\gamma([0,1] \cap (u - \delta, u + \delta)) \subseteq D_\varepsilon(\gamma(u)) \subseteq U$. Choose $u' = \min\{1, u + \delta/2\}$, there is $\gamma(u') \in U$ and thus $\gamma(u') \in U'$. However, $u' > u$ suggests that $f(u') \in V'$, which is a contradiction.

Suppose $\gamma(u) \in V'(\gamma(u) \notin U', u \neq 0)$, then $\gamma(u) \in V$. Since V is open, there exists $\varepsilon > 0$ such that $D_\varepsilon(\gamma(u)) \subseteq V$. By continuity of γ, there exists $\delta > 0$ such that $\gamma([0,1] \cap (u - \delta, u + \delta)) \subseteq D_\varepsilon(\gamma(u)) \subseteq V$. Choose $u' = \max\{0, u - \delta/2\}$. Then, for every $u'' \in [u', 1]$, there is $\gamma(u'') \in V$ and thus $\gamma(u'') \in V'$. However, $u' < u$ and u' is an upper bound of A, which is a contradiction.

Therefore, S must be connected.

■

Note that the converse of Theorem 1.4.14 ("Every connected set is path-connected") is not true. An example is the topologist sine curve.

Example 2.3.5 Let $S = \{(x, \sin(x)) : 0 < x \leq 1\}$ and $S' = \{(0, y) : y \in [-1, 1]\}$ which is the set of limit points of S. Let $A = \overline{S} = S \cup S'$.

A is not path-connected. A path that starts on S is constrained to follow the sine curve, but it can never reach the imaginary axis (for example, $\sin(1/x)$ is not defined at $x = 0$).

A is connected. Both S and S' are path-connected. Since the sine curve gets infinitely close to the imaginary axis (S'), any open set containing S' must intersect with S. So there does not exist

two non-empty disjoint open sets that A is a subset of. Therefore A is connected.

For a rigorous proof, see a text on topology. Other examples of connected by not path connected sets include the deleted infinite comb and the deleted infinite broom.

However, if a connected set is open, it must be path-connected.

Proposition 2.3.6 If $S \subseteq \mathbb{C}$ is open and connected, then S is path-connected.

Proof. Proof to be completed ■

Proposition 2.3.7 If $C \in \mathbb{C}$ is an open connected set and let $a, b \in C$, then there is a differentiable path $\gamma : [0, 1] \to C$ with $\gamma(0) = a$ and $\gamma(1) = b$.

Proof. Proof to be completed ■

Definition 2.3.8 A set $S \subseteq \mathbb{C}$ is a ***complex domain / region*** if and only if it is open and connected.

Proposition 2.3.9 Let $f : C \subseteq \mathbb{C} \to \mathbb{C}$ be a continuous function. If C is connected, then $f(C)$ is connected.

Proof. Proof to be completed ■

2.4 Compact Sets

Definition 2.4.1 Let $S \subseteq \mathbb{C}$. An open cover is a collection of open sets $\{U_\alpha\}_{\alpha \in \mathcal{A}}$ for α in some index set $\mathcal{A}$, such that $S \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha$.
A ***subcover*** is a subset of the collection.

Definition 2.4.2 A set $K \subseteq \mathbb{C}$ is ***compact*** if every open cover of K has a finite subcover, i.e.

$$\text{If } \{U_\alpha\}_{\alpha \in \mathcal{A}} \text{ are open sets s.t. } K \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha, \exists \{\alpha_1, ..., \alpha_n\}, K \subseteq \bigcup_{i=1}^{n} U_{\alpha_i}$$

Theorem 2.4.3 (Heine-Borel) Let $K \subseteq \mathbb{C}$. The following are equivalent:

(i) K is closed and bounded.

(ii) K is compact.

(iii) Every sequence of points in K has a subsequence which converges to some point in K.

Proof. Proof to be completed ■

Theorem 2.4.4 Let $f : K \subset \mathbb{C} \to \mathbb{R}$. If f is continuous and K is a compact set, f attains finite maximum and minimum values.

Proof. Let $f : K \subset \mathbb{C} \to \mathbb{R}$. Assume f is continuous and K is compact. By Proposition 2.4.4, $f(K)$ is compact. By Theorem 2.4.3, $f(K)$ is closed and bounded. Since it is bounded, $M = \sup\{f(z) : z \in K\}$ and $m = \inf\{f(z) : z \in K\}$ are finite. Since $f(K)$ is closed, all limits points of $f(K)$ belong to it. Therefore, $m, M \in f(K)$. ■

Lemma 2.4.5 (Distance Lemma) Suppose $K, C \in \mathbb{C}$, K is compact, C is closed, and $K \cap C = \varnothing$. For all $z \in K, w \in C$, there exists $\rho > 0$ such that $|z - w| > \rho$.

Proof. Consider $C^{\complement} = \mathbb{C} \backslash C$. Since C is closed, $C^{\complement}$ is open. Since $C^{\complement}$ is open, for all points $z_0 \in C^{\complement}$, there exists $\varepsilon > 0$ such that $B_\varepsilon(z_0) \subseteq C^{\complement}$. Since $C^{\complement} \cap C = \varnothing$, the distance between any point in C and any point in $C^{\complement}$ is positive.

$K \subseteq C^{\complement}$. (Fix $a \in K$. Since $K \cap C = \varnothing$, $a \notin C$, so $a \in C^{c}omplement$)

Hence for all $z \in K, w \in C$, there is $\rho_z > 0$ such that $|z - w| > \rho_z$. An open cover for K is the collection of $U_z = B_{\rho_z}(z)$. Also for each z, each point in $B_{\rho_z}(z) \cap K$ is at least ρ_z from C. Since K is compact, the open cover has a finite subcover $U_{z_1}, U_{z_2}, ..., U_{z_n}$. Choose $\rho = \min\{\rho_{z_1}, \rho_{z_2}, ..., \rho_{z_n}\}$. ρ satisfies $|z - w| > \rho$ for all $z \in K, w \in C$. ■

Proposition 2.4.6 Let $f : K \to \mathbb{C}$ where K is compact. $f(K)$ is also compact.

Proof. Let $C = \{U_\alpha\}_{\alpha \in \mathcal{A}}$ be an open cover of $f(K)$. Then $f^{-1}(C)$ is an open cover of K. Since K is compact, C has a finite subcover $f^{-1}(U_{\alpha_1}) \cup f^{-1}(U_{\alpha_2}) \cup ... \cup f^{-1}(U_{\alpha_n})$ and $K \subset f^{-1}(U_{\alpha_1}) \cup f^{-1}(U_{\alpha_2}) \cup ... \cup f^{-1}(U_{\alpha_n})$. Therefore $f(K) \subset U_{\alpha_1} \cup ... \cup U_{\alpha_n}$. By definition, $f(K)$ is compact. ■

Proposition 2.4.7 If $f : K \to \mathbb{C}$ is continuous and $K \subset \mathbb{C}$ is compact, then f is uniformly continuous.

Proof. Let $K \subset \mathbb{C}$ be a compact set. Let $f : K \to \mathbb{C}$ be a continuous function. Fix $\varepsilon > 0$.

Since f is continuous, for all points $t \in K$, there exists δ_t such that $|s - t| < \delta_t \implies |f(s) - f(t)| < \frac{\varepsilon}{2}$.

An open cover of K is the $\{B_{\delta/2}(z) : z \in K\}$. Since K is compact, it has finite subcover $\{B_{\delta/2}(z_n) : z_n \in K, n = 1, 2, ..., m\}$. Choose $\delta = \min\{\frac{\delta_{z_n}}{2} : n = 1, 2, ..., m\}$ and suppose for some $s, x \in K$, $|s - x| < \delta$. Since $\{B_{\delta/2}(z_n) : z_n \in K, n = 1, 2, ..., m\}$ is an open over, $s \in B_{\delta_{z_i}/2}(z_i)$ for some $i \in \{1, ..., m\}$. So we have $|f(s) - f(z_i)| < \frac{\varepsilon}{2}$.

Since $|s - x| < \delta \leq \frac{\delta_{z_i}}{2}$, $|x - z_i| \leq |s - x| + |s - z_i| < \frac{\delta_{z_i}}{2} + \frac{\delta_{z_i}}{2} = \delta_{z_i}$ so $x \in B_{\delta_{z_i}}(z_i)$. So we have $|f(x) - f(z_i)| < \frac{\varepsilon}{2}$.

$$|f(s) - f(x)| \leq |f(s) - f(z_i)| + |f(x) - f(z_i)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Therefore, f is uniformly continuous on K. ■

Lemma 2.4.8 Suppose $\gamma : [a, b] \to G$ is a continuous path from the interval $[a, b]$ into an open set $G \subseteq \mathbb{C}$. Then there are a number $\rho > 0$ and a subdivision of the interval $a = t_0 < t_1 < t_2 < ... < t_n = b$ such that

(i) $B_\rho(\gamma(t_k)) \subset G$ for all k

(ii) $\gamma(t) \in B_\rho(\gamma(t_0))$ for $t_0 \leq t \leq t_1$

(iii) $\gamma(t) \in B_\rho(\gamma(t_k))$ for $t_{k-1} \leq t \leq t_{k+1}$

(iv) $\gamma(t) \in B_\rho(\gamma(t_n))$ for $t_{n-1} \leq t \leq t_n$

Proof. Let $G \subseteq \mathbb{C}$ be an open set. Let $\gamma : [a, b] \to G$ be continuous. Since γ is continuous and $[a, b]$ is compact, by proposition 2.4.6, $\gamma([a, b])$ is compact.

Since $\gamma([a, b]) \subseteq G$, $\gamma([a, b]) \cap G^{\complement} = \varnothing$. Also $\gamma([a, b])$ is compact, $G^{\complement}$ is closed (G is open). By Lemma 2.4.5, there exists $\rho > 0$ such that for every point $t \in \gamma([a, b])$ $B_\rho(\gamma(t)) \subset G$. ($\implies$ (i))

Since γ is continuous on compact set $[a, b]$, by Proposition 2.4.7, γ is uniformly continuous.

By definition, there exists $\delta > 0$ such that $|s - t| < \delta \implies |\gamma(t) - \gamma(s)| < \rho$. Choose ρ and the subdivision of the interval such that $t_k - t_{k-1} < \delta$ for all $k = 1, 2, 3, ..., n$ ($\implies$ (ii), (iii), (iv)) ■

3 Complex Differentiation

3.1 The Complex Derivative

Definition 3.1.1 Let $f : A \to \mathbb{C}$ where $A \subseteq \mathbb{C}$ is an open set. f is ***complex differentiable*** at $z_0 \in A$ if

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \quad \text{exists}$$

f is ***holomorphic*** on A if f is complex differentiable at each $z_0 \in A$.

As in singe variable calculus, complex differentiable implies continuity.

Proposition 3.1.2 Let $f : A \to \mathbb{C}$ where $A \subseteq \mathbb{C}$ is an open set and $z_0 \in A$. If f is complex differentiable at z_0, f is continuous at z_0.

Proof. Let $f : A \to \mathbb{C}$. Suppose f is complex differentiable at z_0, then

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \quad \text{exists}$$

To show f is continuous at z_0, we would like to show that $\lim_{z \to z_0} f(z) - f(z_0) = 0$.

$$\begin{aligned}
\lim_{z \to z_0} [f(z) - f(z_0)] &= \lim_{z \to z_0} \left[\frac{f(z) - f(z_0)}{z - z_0} (z - z_0) \right] \\
&= \lim_{z \to z_0} \left[\frac{f(z) - f(z_0)}{z - z_0} \right] \cdot \lim_{z \to z_0} (z - z_0) \\
&= \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \cdot 0 = 0
\end{aligned}$$

Thus $\lim_{z\to z_0}[f(z)-f(z_0)]=0 \Longrightarrow \lim_{z\to z_0} f(z)=f(z_0)$. ■

The usual rules of calculus – the product rule, the quotient rule and the chain rule also apply to holomorphic functions.

Proposition 3.1.3 Let $f : A \to \mathbb{C}$ and $g : A \to \mathbb{C}$ and $A \subseteq \mathbb{C}$. Let $a, b \in \mathbb{C}$. If f, g are complex differentiable at $z_0 \in A$

(i) $af + bg$ is complex differentiable at z_0 and

$$(af+bg)'(z_0) = af'(z) + bg'(z_0)$$

(ii) fg is complex differentiable at z_0 and

$$(fg)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0)$$

(iii) If $g(z_0) \neq 0$, the f/g is complex differetiable at z_0 and

$$\left(\frac{f}{g}\right)'(z_0) = \frac{f'(z)g(z_0) - g'(z)f(z_0)}{[g(z_0)]^2}$$

Proof. Let $f : A \to \mathbb{C}$ and $g : A \to \mathbb{C}$, and $\lim_{z\to z_0} f(z) = f(z_0)$

(i) By limit properties (Proposition 2.2.3),

$$\begin{aligned}
\lim_{z\to z_0} \frac{[af(z)+bg(z)]-[af(z_0)+g(z_0)]}{z-z_0} &= \lim_{z\to z_0}\left[\frac{af(z)-af(z_0)}{z-z_0} + \frac{bg(z)-bg(z_0)}{z-z_0}\right] \\
&= \lim_{z\to z_0}\left[\frac{af(z)-af(z_0)}{z-z_0}\right] + \lim_{z\to z_0}\left[\frac{bg(z)-bg(z_0)}{z-z_0}\right] \\
&= a\cdot\lim_{z\to z_0}\left[\frac{f(z)-f(z_0)}{z-z_0}\right] + b\cdot\lim_{z\to z_0}\left[\frac{g(z)-g(z_0)}{z-z_0}\right] \\
&= af'(z_0) + bg'(z_0) \quad \text{by definition}
\end{aligned}$$

(ii) By limit properties (Proposition 2.2.3),

$$\begin{aligned}
\lim_{z\to z_0} \frac{f(z)g(z)-f(z_0)g(z_0)}{z-z_0} &= \lim_{z\to z_0}\left[\frac{f(z)g(z)-f(z)g(z_0)}{z-z_0} + \frac{f(z)g(z_0)-f(z_0)g(z_0)}{z-z_0}\right] \\
&= \lim_{z\to z_0}\left[f(z)\frac{g(z)-g(z_0)}{z-z_0}\right] + \lim_{z\to z_0}\left[\frac{f(z)-f(z_0)}{z-z_0}g(z_0)\right] \\
&= \lim_{z\to z_0} f(z)\cdot\lim_{z\to z_0}\left[\frac{g(z)-g(z_0)}{z-z_0}\right] + \lim_{z\to z_0}\left[\frac{f(z)-f(z_0)}{z-z_0}\right]\cdot\lim_{z\to z_0} g(z_0) \\
&= f(z_0)g'(z_0) + f'(z_0)g(z_0) \quad \text{by definition}
\end{aligned}$$

(iii) We first need the reciprocal rule: $(1/f)'(z) = -f'(z)/[f(z)]^2$.

$$\begin{aligned}
\left(\frac{1}{f}\right)'(z) &= \lim_{z\to z_0} \frac{(1/f)(z) - (1/f)(z_0)}{z - z_0} = \lim_{z\to z_0} \frac{(1/f(z)) - (1/f(z_0))}{z - z_0} \\
&= \lim_{z\to z_0} \frac{\frac{f(z_0)-f(z)}{(f(z_0)f(z))^2}}{z - z_0} \\
&= \lim_{z\to z_0} \frac{f(z_0) - f(z)}{z - z_0} \frac{1}{(f(z_0)f(z))^2} = \lim_{z\to z_0} \frac{f(z_0) - f(z)}{z - z_0} \cdot \lim_{z\to z_0} \frac{1}{(f(z_0)f(z))^2} \\
&= -f'(z_0)/[f(z_0)]^2
\end{aligned}$$

Then we will prove the quotient rule,

$$\begin{aligned}
\left(\frac{f}{g}\right)'(z) &= (f \cdot \frac{1}{g}) \\
&= f' \cdot \left(\frac{1}{g}\right) + f \cdot \left(\frac{1}{g}\right)' \quad \text{by product rule} \\
&= f' \cdot \left(\frac{1}{g}\right) + f \cdot -\frac{g'}{g^2} \quad \text{by reciprocal rule} \\
&= \frac{f'(z_0)g(z_0) - g'(z_0)f(z_0)}{[g(z_0)]^2}
\end{aligned}$$

■

Proposition 3.1.4 The following are true:

(i) Any polynomial $a_0 + a_1 z + ... + a_n z^n$ is holomorphic on all $\mathbb{C}$ with derivative $a_1 + 2a_2 z + ... + na_n z^{n-1}$.

(ii) Any rational function

$$\frac{a_0 + a_1 z + ... + a_n z^n}{b_0 + b_1 z + ... + b_m z^m}$$

is holomorphic on the open set consisting of all z except those (at most, m) points where the denominator is zero.

Proof. (i) By Proposition 3.1.3 (i), it suffices to show that each term in the polynomial $a_0 + a_1 z + ... + a_n z^n$ is holomorphic on all $\mathbb{C}$. Proof to be completed

■

Theorem 3.1.5 (Chain Rule) Let $f : A \to \mathbb{C}$ and $g : B \to \mathbb{C}$ be holomorphic (A, B are open sets) and $f(A) \subseteq B$. Then $g \circ f : A \to \mathbb{C}$ is holomorphic and

$$(g \circ f)'(z) = g'(f(z)) \cdot f'(z)$$

Proof. Let $f : A \to \mathbb{C}$ and $g : B \to \mathbb{C}$ be holomorphic and $f(A) \subseteq B$. Let $w_0 = f(z_0)$ and define $h : B \to \mathbb{C}$ to be

$$h(w) = \begin{cases} \frac{g(w)-g(w_0)}{w-w_0} - g'(w_0) & \text{if } w \neq w_0 \\ 0 & \text{if } w = w_0 \end{cases}$$

Since $g'(w_0)$ exists (it is holomorphic on B), h is continuous. Then by Theorem 2.2.6,

$$\lim_{z \to z_0} h(f(z)) = h(w_0) = 0$$

Let $w = f(z)$. By the definition of $h(w)$, there is $g(f(z)) - g(w_0) = [h(f(z)) + g'(w_0)][f(z) - w_0]$. For $z \neq z_0$,

$$\begin{aligned} \frac{g(f(z)) - g(f(z_0))}{z - z_0} &= [h(f(z)) + g'(w_0)]\frac{f(z) - w_0}{z - z_0} \\ \lim_{z \to z_0} \frac{g(f(z)) - g(f(z_0))}{z - z_0} &= \lim_{z \to z_0} [h(f(z)) + g'(w_0)]\frac{f(z) - w_0}{z - z_0} \\ &= \lim_{z \to z_0} [h(f(z)) + g'(w_0)] \cdot \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \\ &= [0 + g'(w_0)] \cdot f'(z_0) = g'(f(z_0)) \cdot f'(z_0) \end{aligned}$$

■

Proposition 3.1.6 Let $A \subseteq \mathbb{C}$ be open and connected and let $f : A \to \mathbb{C}$ be holomorphic on A. If $f'(z) = 0$ on A for all $z \in A$, then f is constant on A.

Proof. Let $A \subseteq \mathbb{C}$ be open and connected. By Proposition 2.3.6, A is connected. Fix $a, b \in A$. There exists a continuous function $\gamma : [0, 1] \to \mathbb{C}$ such that $\gamma(0) = a$ and $\gamma(1) = b$. Let $u = \mathrm{Re}(f), v = \mathrm{Im}(f)$. By chain rule, $(f \circ \gamma)'(t) = f'(\gamma(t))\gamma'(t)$. Since $f'(z) = 0$ for all $z \in A$, $f'(\gamma(t))\gamma'(t) = 0$. Since

$$\begin{aligned} (f \circ \gamma)'(t) &= \lim_{x \to t} \frac{(f \circ \gamma)(x) - (f \circ \gamma)(t)}{x - t} \\ &= \lim_{x \to t} \mathrm{Re}\left(\frac{(f \circ \gamma)(x) - (f \circ \gamma)(t)}{x - t}\right) + \lim_{x \to t} \mathrm{Im}\left(\frac{(f \circ \gamma)(x) - (f \circ \gamma)(t)}{x - t}\right) i \\ &= \lim_{x \to t} \frac{(u \circ \gamma)(x) - (u \circ \gamma)(t)}{x - t} + \lim_{x \to t} \frac{(v \circ \gamma)(x) - (v \circ \gamma)(t)}{x - t} i \\ &= (u \circ \gamma)'(t) + (v \circ \gamma)'(t)i \end{aligned}$$

$(u \circ \gamma)'(t) = 0$ and $(v \circ \gamma)'(t)i = 0$. By mean value theorem in calculus, $u \circ \gamma = j$ for some $j \in \mathbb{R}$ and $v \circ \gamma = k$ for some $k \in \mathbb{R}$. Therefore $\mathrm{Re}(a) = \mathrm{Re}(b)$ and $\mathrm{Im}(a) = \mathrm{Im}(b)$ for arbitrary $a, b \in A$. ■

3.2 Conformal Maps

Informally, conformal maps are functions on $\mathbb{C}$ that preserve angles between curves. More precisely, suppose f is differentiable at z_0, then the function maps z_0 to $f(z_0)$ and the curve $\gamma(t)$ to $f(\gamma(t))$. Under this map, the tangent vector $\gamma'(t_0)$ at z_0 is mapped to the tangent vector $f'(\gamma(t_0))$ at $f(z_0)$. Here is a more formal definition.

Definition 3.2.1 A map $f : A \subseteq \mathbb{C} \to \mathbb{C}$ is ***conformal*** at z_0 if there exists $\theta \in [0, 2\pi)$ and $r > 0$ such that for any curve $\gamma(t)$ for which $\gamma(t) \in A$, $\gamma(t_0) = z_0$, γ differentiable at t_0 and $\gamma'(t_0) \neq 0$, the curve $\sigma(t) = f(\gamma(t))$ is differentiable at t_0 and, setting $u = \sigma'(t_0)$ and $v = \gamma'(t_0)$, we have $|u| = r|v|$ and $\arg u = \arg v + \theta + 2k\pi$ for some $k \in \mathbb{Z}^+$. A map is a ***conformal map*** when it is conformal at every point.

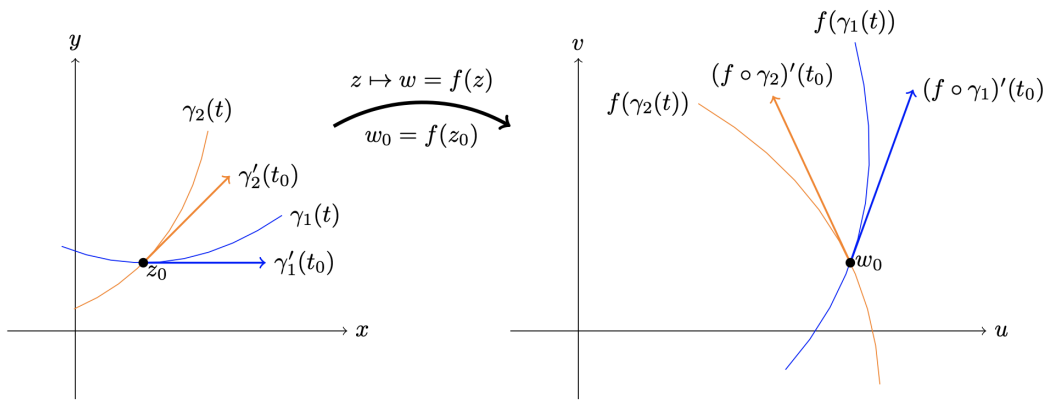

Figure 1: The figure below a conformal map $f(z)$ mapping two curves through z_0 to two curves through $w_0 = f(z_0)$. The tangent vectors to each of the original curves are both rotated and scaled by the same amount. A conformal map rotates and scales all tangent vectors at z0 by the same amount.

Theorem 3.2.2 (Conformal Mapping Theorem) If $f : A \subseteq \mathbb{C}$ is holomorphic at z_0 and $f'(z_0) \neq 0$, then f is conformal at z_0 with $\theta = \arg f'(z_0)$ and $r = |f'(z_0)|$.

Proof. Using the preceding notation, $u = \sigma'(0) = f'(\gamma(0)) \cdot \gamma'(0) = f'(z_0) \cdot v$. Since $u, f'(z_0), v \in \mathbb{C}$, by Proposition 1.2.1, $\arg u = \arg f'(z_0) + \arg v + 2k\pi$ for some $k \in \mathbb{Z}$ and $|u| = |f'(z_0)| \cdot |v|$. ■

3.3 Cauchy-Riemann Equations

Theorem 3.3.1 (Cauchy-Riemann Theorem) Let $f : A \subseteq \mathbb{C} \to \mathbb{C}$ where A is an open set, and $f(x, y) = u(x, y) + iv(x, y)$. Let $z_0 = (x_0, y_0) \in A$. f is complex differentiable at z_0 if and only if its real and imaginary parts are real differentiable at z_0 and at z_0,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Note that thus if the partials above exist and are continuous on A, and satisfy the Cauchy-Riemann equations, then f is holomorphic on A.

Proof. ($\Longrightarrow$):

Suppose $f'(z_0)$ exists, we want to show that its real and imaginary parts are real differentiable at z_0 and satisfy the Cauchy-Riemann equations at z_0. We will have to compute $f'(z_0)$ from the

real and imaginary directions. In the limit

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

we first consider the case where we approach z_0 along the imaginary direction. Set $z = x + iy_0$.

$$\begin{aligned}
\frac{f(z) - f(z_0)}{z - z_0} &= \frac{u(x, y_0) + iv(x, y_0) - u(x_0, y_0) - iv(x_0, y_0)}{(x, y_0) - (x_0, y_0)} \\
&= \frac{u(x, y_0) - u(x_0, y_0)}{x - x_0} + i\frac{v(x, y_0) - v(x_0, y_0)}{x - x_0} \\
\lim_{x \to x_0} \frac{f(z) - f(z_0)}{z - z_0} &= \lim_{x \to x_0} \frac{u(x, y_0) - u(x_0, y_0)}{x - x_0} + i \lim_{x \to x_0} \frac{v(x, y_0) - v(x_0, y_0)}{x - x_0} \\
&= \left.\frac{\partial u}{\partial x}\right|_{(x_0, y_0)} + i\left.\frac{\partial v}{\partial x}\right|_{(x_0, y_0)} \quad \text{since by Proposition 2.2.9, both limits exist}
\end{aligned}$$

Now consier the case where we approach z_0 along the real direction. Set $z = x_0 + iy$.

$$\begin{aligned}
\frac{f(z) - f(z_0)}{z - z_0} &= \frac{u(x_0, y) + iv(x_0, y) - u(x_0, y_0) - iv(x_0, y_0)}{(x_0, y) - (x_0, y_0)} \\
&= \frac{u(x_0, y) - u(x_0, y_0)}{i(y - y_0)} + i\frac{v(x_0, y) - v(x_0, y_0)}{i(y - y_0)} \\
\lim_{y \to y_0} \frac{f(z) - f(z_0)}{z - z_0} &= -i \lim_{y \to y_0} \frac{u(x_0, y) - u(x_0, y_0)}{y - y_0} + \lim_{y \to y_0} \frac{v(x_0, y) - v(x_0, y_0)}{(y - y_0)} \\
&= -i\left.\frac{\partial u}{\partial y}\right|_{(x_0, y_0)} + \left.\frac{\partial v}{\partial y}\right|_{(x_0, y_0)} \quad \text{since by Proposition 2.2.9, both limits exist}
\end{aligned}$$

Since $f'(z_0)$ exists and has the same value regardless of which direction z approaches z_0, both limits are equal, and we have

$$f'(z_0) = \left.\frac{\partial u}{\partial x}\right|_{(x_0, y_0)} + i\left.\frac{\partial v}{\partial x}\right|_{(x_0, y_0)} = \left.\frac{\partial v}{\partial y}\right|_{(x_0, y_0)} - i\left.\frac{\partial u}{\partial y}\right|_{(x_0, y_0)}$$

Comparing the real and imaginary parts,

$$\left.\frac{\partial u}{\partial x}\right|_{(x_0, y_0)} = \left.\frac{\partial v}{\partial y}\right|_{(x_0, y_0)} \quad \text{and} \quad \left.\frac{\partial v}{\partial x}\right|_{(x_0, y_0)} = -\left.\frac{\partial u}{\partial y}\right|_{(x_0, y_0)}$$

($\Longleftarrow$):

Take f to be a function from $\mathbb{R}^2$ to $\mathbb{R}^2$. Then there exists a linear map $L : \mathbb{R}^2 \to \mathbb{R}^2$ such that for all $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |z - z_0| < \delta \Longrightarrow \frac{|f(z) - f(z_0) - L(z - z_0)|}{z - z_0} < \varepsilon$$

and the linear map L is the differential of f at z_0. Since f is (real) differentiable at z_0, the matrix of a the differential[2] is the Jacobian matrix of f at z_0, where the Jacobian is given by

$$\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}$$

[2]Every linear transformation can be represented by a matrix, see a text on calculus for the proof

By Proposition 1.2.2, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ to represent complex multiplication, thus we have obtained the Cauchy-Riemann equations. ■

A twice continuously differentiable function $u : A \to \mathbb{R}$ defined on an open set A is called ***harmonic*** if

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

where the expression $\nabla^2 u$ is called the ***Laplacian***. The Cauchy Riemann equations can be used to show that the real and imaginary parts of a complex function are harmonic.

Proposition 3.3.2 Let $f : A \subseteq \mathbb{C} \to \mathbb{C}$ and A is an open set. If f is holomorphic on A, and $f = u + iv$, then u and v are harmonic on A.

Proof. Let $f : A \subseteq \mathbb{C} \to \mathbb{C}$ and $f = u+iv$. By the Cauchy-Riemann theorem, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. Therefore $\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y}$ and $\frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x}$. Since f is holomorphic, by Clairaut's theorem, $-\frac{\partial^2 v}{\partial x \partial y} = -\frac{\partial^2 v}{\partial y \partial x}$. Therefore $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. Similarly, $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$. ■